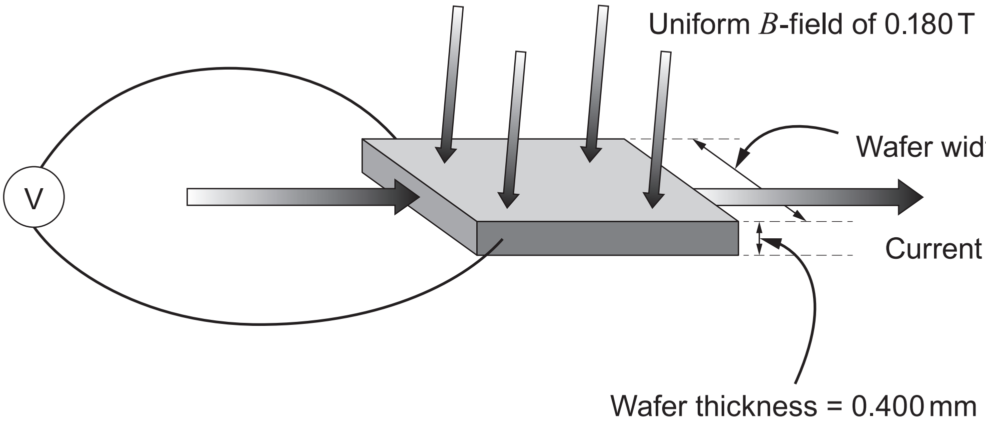
Examiner only

6. (a) Catrin carries out a Hall effect experiment to find out the number of free electrons per unit volume in a certain metal. She places a wafer of the metal in a known magnetic field and passes a current through it (see diagram).



(i) By considering the forces acting on free electrons passing through the wafer, explain why a Hall voltage is measured on the voltmeter shown. [4]

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(ii) By equating the magnetic and electrical forces, show that the Hall voltage, V_H, is given by:

$$V_H = Bvd$$

where B is the magnetic flux density, d is the width of the wafer and v is the drift velocity. [3]

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Examiner only

(iii) Catrin states that her measured Hall voltage of 68.0 nV is consistent with a drift velocity of approximately 0.07 mm s^{-1}. Determine whether, or not, she is correct. [2]

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(iv) Calculate the number of free electrons per unit volume for the metal. [3]

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(b) After repeating the same experiment on a different metal, Catrin obtains a value of $(5.85 \pm 0.19) \times 10^{28}\,\text{m}^{-3}$, for the number of free electrons per unit volume. She is given a table of values in order to determine which metal has been used in the experiment.

Element	Free electron density / $10^{22}\,\text{cm}^{-3}$
Aluminium	18.1
Barium	3.15
Copper	8.47
Gold	5.90
Iron	17.0
Silver	5.86

Explain which metal(s) she should conclude has been used in the experiment. [2]

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END OF PAPER